

Data Engineering Projects

Project 1: ETL Pipeline with Open Data (CSV -> Parquet -> BigQuery)

Project Overview:

Build a modern ETL pipeline that ingests open CSV data, cleans and transforms it using efficient tools (like Polars), stores in Parquet format, and loads into Google BigQuery for querying.

Milestone 1: Data Collection, Exploration & Preprocessing

- Collect raw open data (e.g., weather or transportation CSV).
- Explore schemas and quality issues.
- Convert CSV to Parquet for efficient storage.

Milestone 2: Model / System Development & Evaluation

- Develop transformation workflows in Python (Polars/Pandas).
- Validate data correctness and completeness.

Milestone 3: Deployment (Real-Time or Batch)

- Deploy batch pipeline to run on schedule (cron/Airflow).
- Load Parquet data into BigQuery.

Milestone 4: MLOps / Monitoring / Automation

- Automate pipeline triggers.
- Monitor job success and data freshness.

Milestone 5: Final Documentation, Demo & Presentation

- Document ETL logic, schemas, and BigQuery tables.
- Demo queries and pipeline execution.

Project 2: Weather Data Pipeline with Python and PostgreSQL

Project Overview:

Extract weather data from an API, clean it, and store it in a PostgreSQL database. Containerize services with Docker for portability.

Milestone 1: Data Collection, Exploration & Preprocessing

- Call Open-Meteo API to collect JSON data.
- Normalize and clean JSON fields.

Milestone 2: Model / System Development & Evaluation

- Design PostgreSQL schema.
- Write ETL scripts to insert into PostgreSQL.

Milestone 3: Deployment (Real-Time or Batch)

- Use Docker to run PostgreSQL and Python scripts.

Milestone 4: MLOps / Monitoring / Automation

- Schedule daily pipeline runs.
- Track failures and logs.

Milestone 5: Final Documentation, Demo & Presentation

- Final documentation on API endpoints, schema design, and usage examples.